

1. Which function produced this compact view of variable names, R classes, and a few values? (referring to image)

```
Rows: 250
Columns: 6
$ patient_id <chr> "P001", "P002", "P003", "P004"
$ age <dbl> 34, 45, 28, 52, 61, 39
$ smoker <dbl> 1, 0, 1, 0, 0, 1
$ district <chr> "North", "South", "East"
$ bp_systolic <dbl> 120, 135, 118, 142, 155
$ status <fct> Mild, Severe, Mild, Moderate
```

a) ggplot()

b) mean()

c) glimpse() ✓

d) tabyl()

2. A variable called Smoker is shown in R as dbl with values 1, 0, 1, 0. Which description is appropriate?

Rows: 12		Columns: 4		
	\$ id <dbl>	\$ Smoker <dbl>	\$ age <dbl>	\$ sex <chr>
1	1	1	34	"F"
2	2	0	28	"M"
3	3	1	41	"F"
4	4	0	22	"F"
5	5	0	36	"M"
6	6	1	29	"M"
7	7	0	55	"F"
8	8	1	31	"M"
9	9	0	24	"F"
10	10	0	47	"M"
11	11	1	39	"F"
12	12	0	26	"M"

a) It is a nominal categorical variable. Stored as a numeric double type in R. ✓

b) It is a continuous numeric variable. Stored as a duplicate type in R.

3. A survey records age in completed years. Which code chunk would best visualize the distribution of this numeric discrete variable?

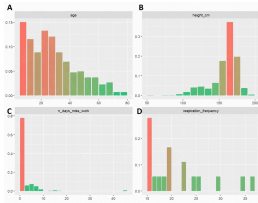
df	
id	age
1	18
2	18
3	19
4	20
5	20
6	20
7	21
8	21
9	22

a) ggplot(df, aes(x = age)) +
geom_histogram(binwidth = 1) ✓

b) ggplot(df, aes(x = age, y = id)) +
geom_col()

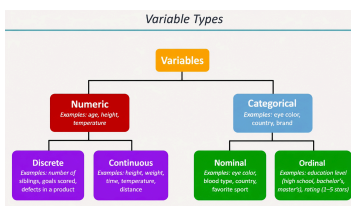
c) ggplot(df, aes(x = age, y = id)) +
geom_line()

4. In the inspect_num output, for which variable might we expect to have the mean median? (referring to image)



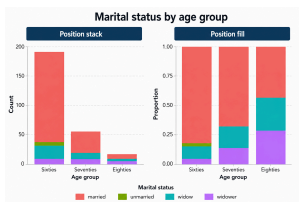
- a) B ✓
- b) C
- c) A
- d) D

5. In the variable-types flowchart, for which branch is a mean least likely to be mea (referring to image)



- a) Numeric → continuous
- b) Categorical → nominal ✓
- c) Categorical → ordinal
- d) Numeric → discrete

6. Why use position = "fill" with ggplots geom_bar or geom_col?



- a) To accentuate the smaller group
- b) To focus on continuous variables within each group
- c) To compare proportions across groups ✓